

Organic Reaction Mechanisms – Aromatic substitutions reactions

CHE 1203 – Organic Chemistry 1

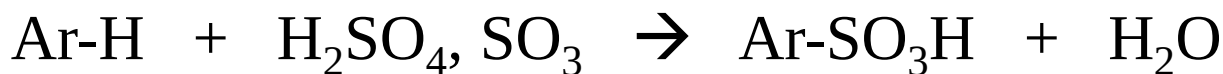
Electrophilic Aromatic Substitution (Aromatic compounds)

Ar-H = aromatic compound

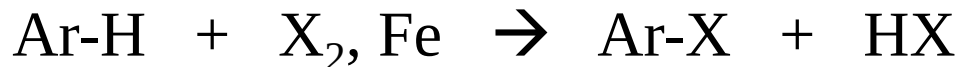
1. Nitration



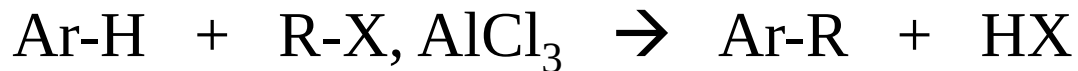
2. Sulfonation

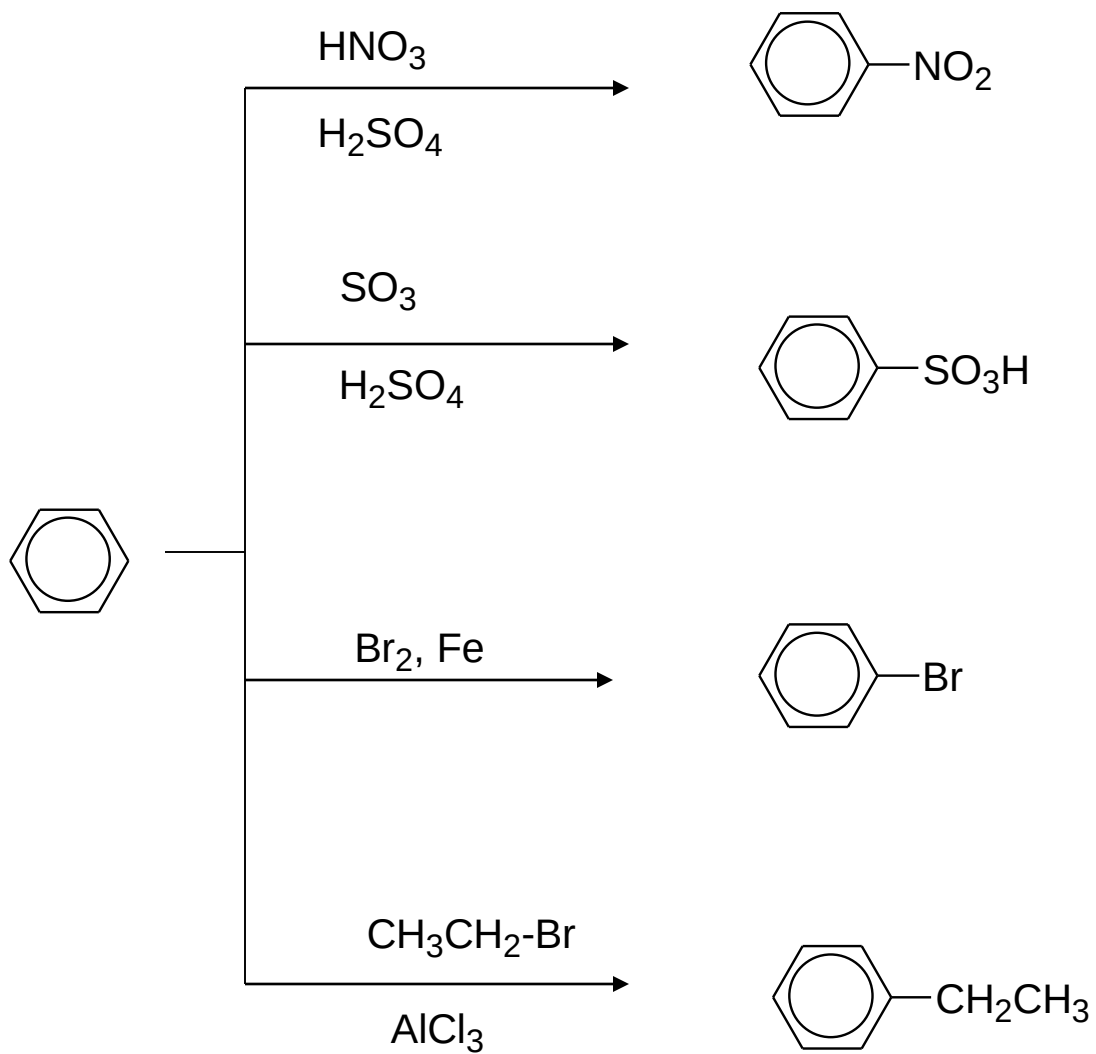


3. Halogenation

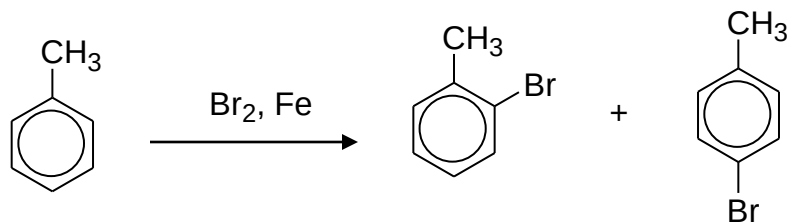
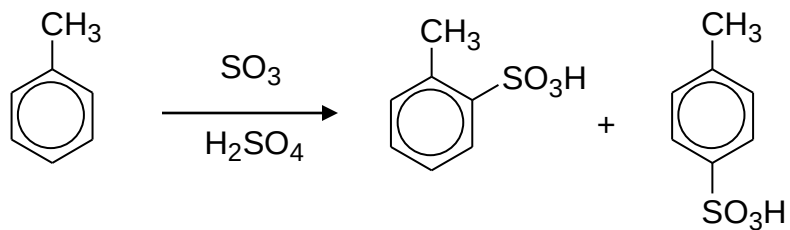
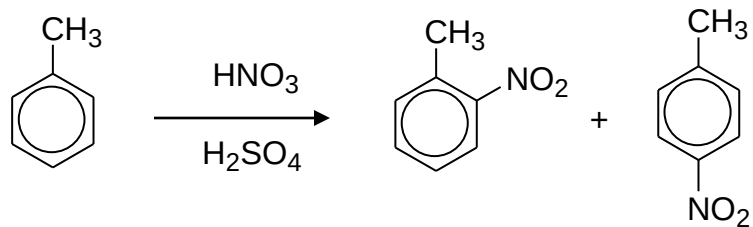


4. Friedel-Crafts alkylation



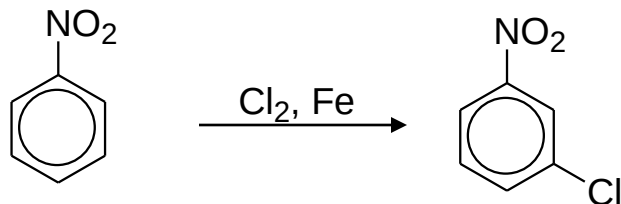
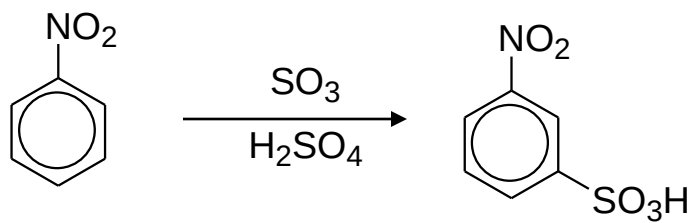
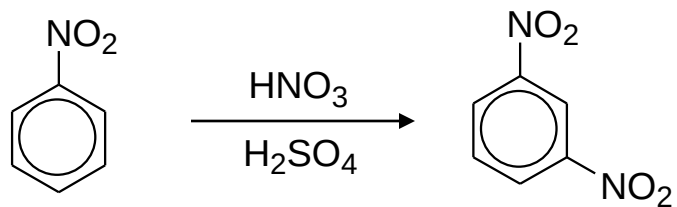


toluene



faster than the same
reactions with
benzene

nitrobenzene



slower than the same
reactions with
benzene

Substituent groups on a benzene ring affect electrophilic aromatic substitution reactions in two ways:

1) reactivity

activate (faster than benzene) or deactivate (slower than benzene)

2) orientation

ortho- + *para*- direction

or *meta*- direction



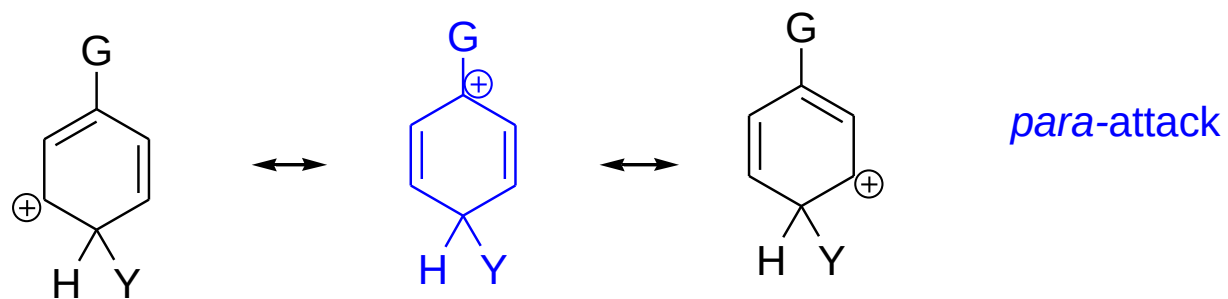
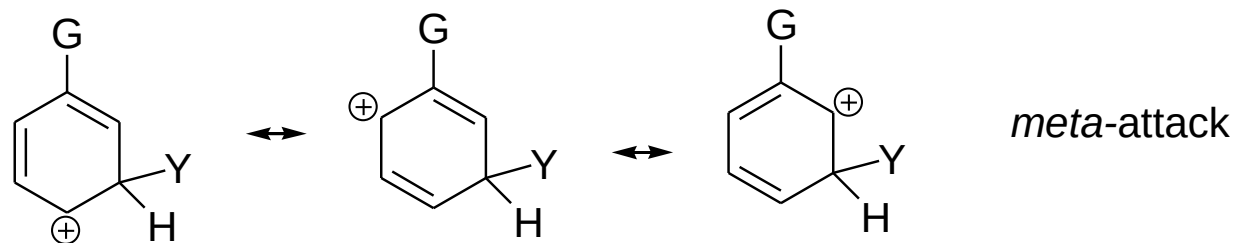
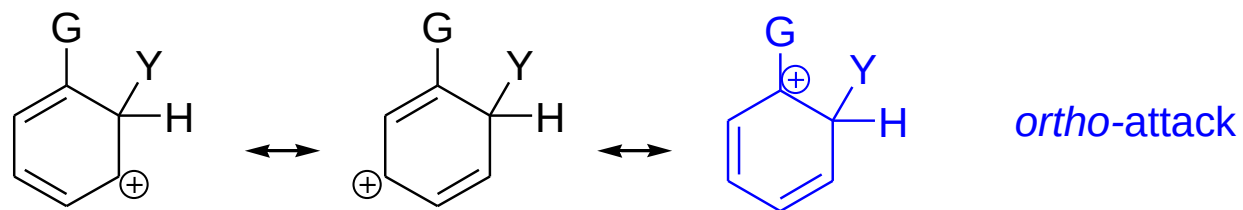
activates the benzene ring towards EAS

directs substitution to the *ortho*- & *para*- positions



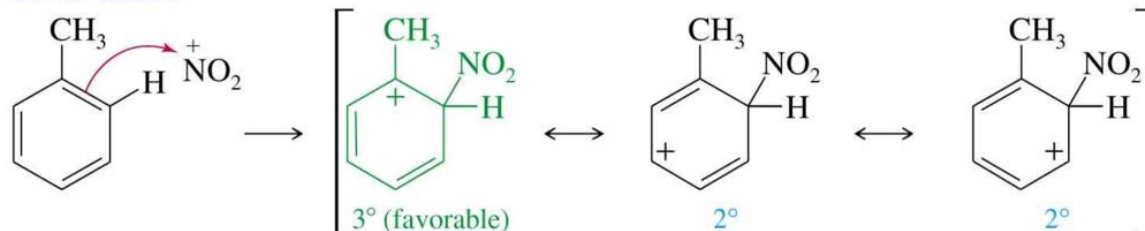
deactivates the benzene ring towards EAS

directs substitution to the *meta*- position

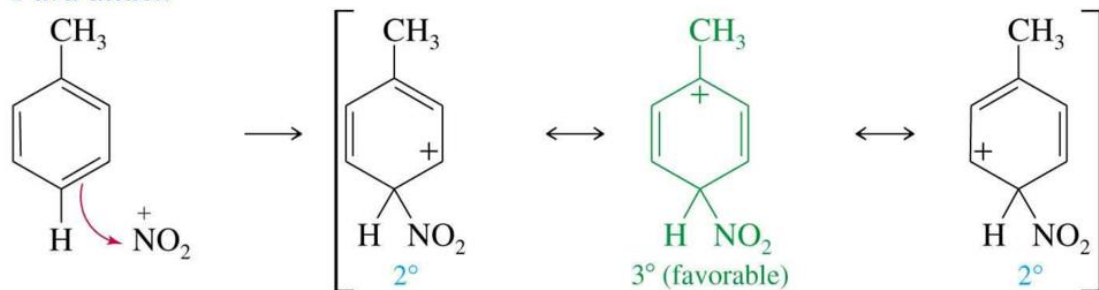


Sigma Complex

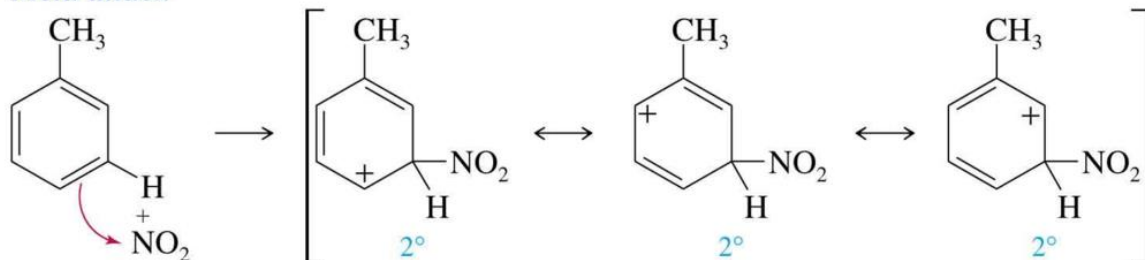
Ortho attack



Para attack



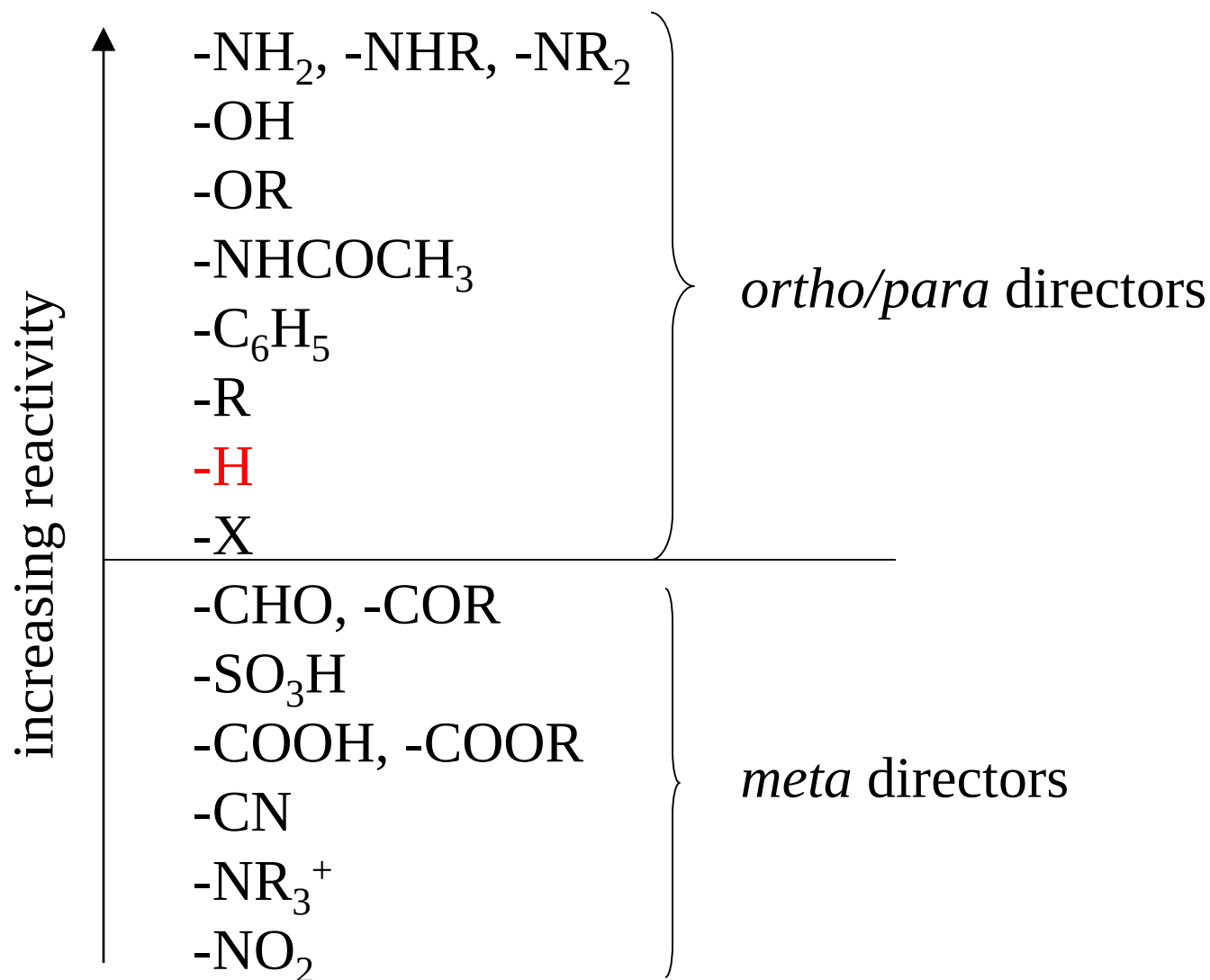
Meta attack

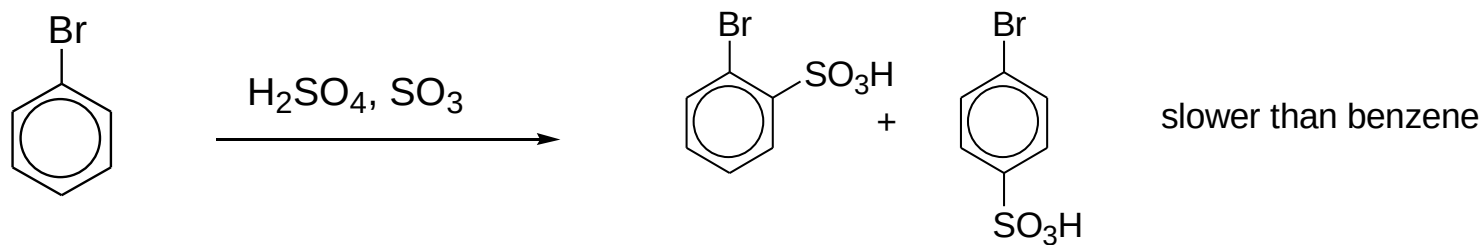
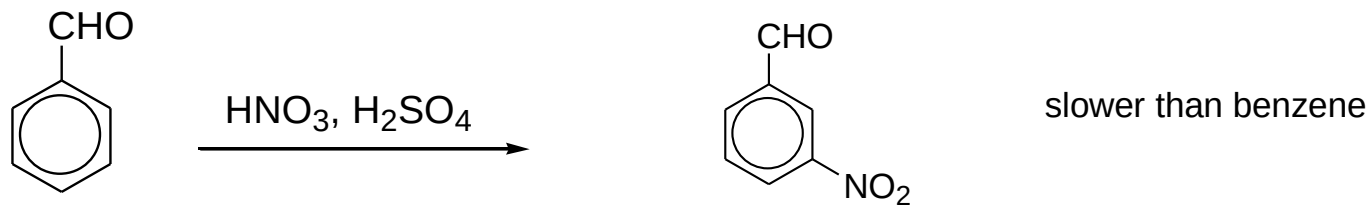
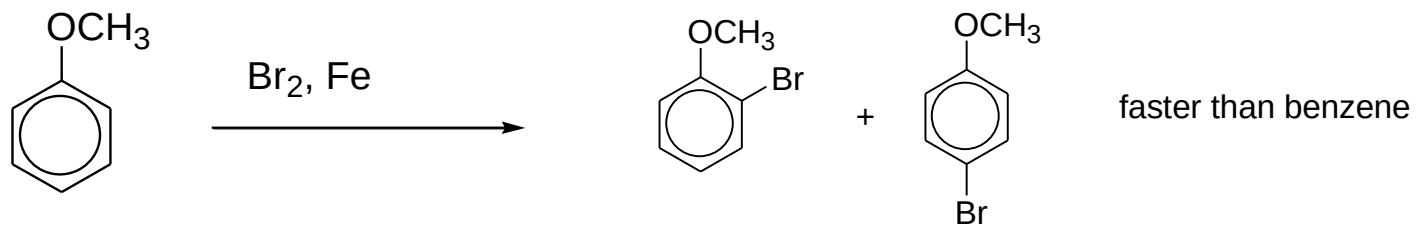


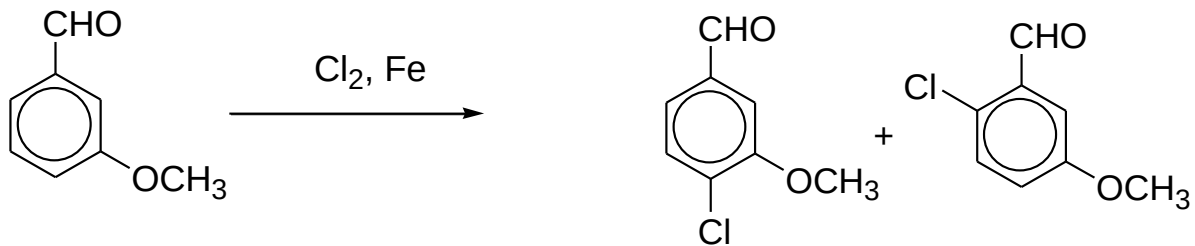
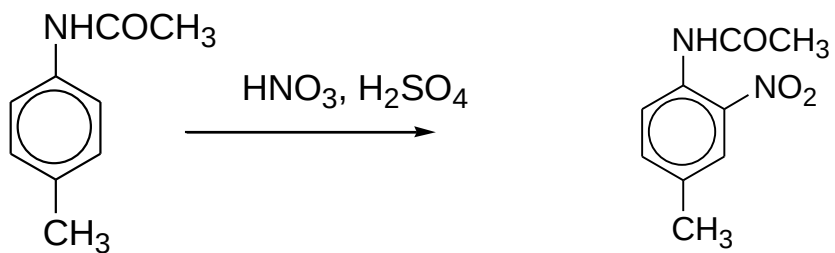
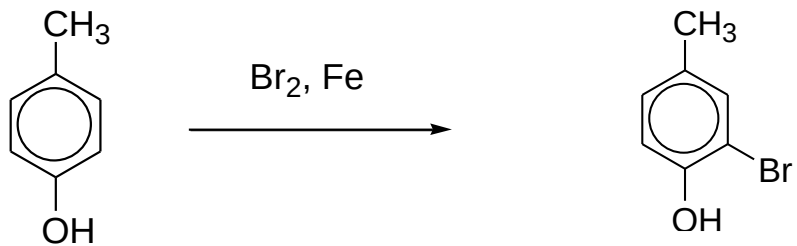
Intermediate is more stable if nitration occurs at the *ortho* or *para* position.

=>

Common substituent groups and their effect on EAS:

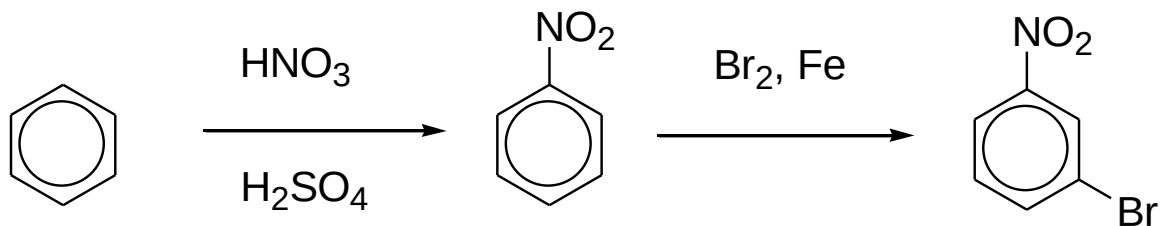




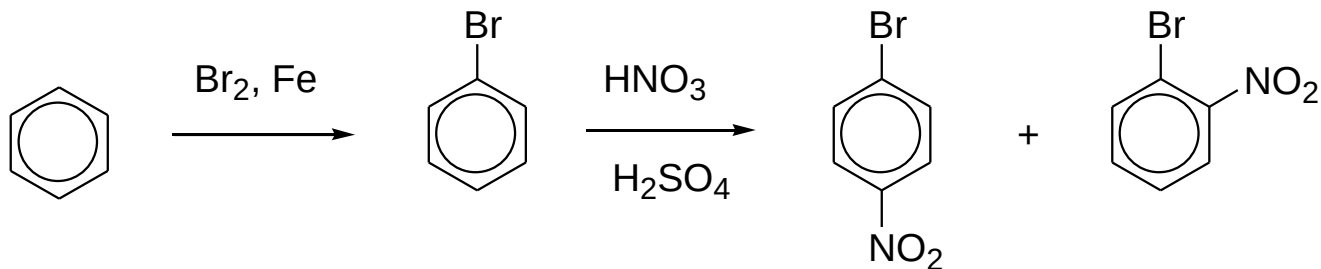


Orientation and synthesis. Order is important!

synthesis of *m*-bromonitrobenzene from benzene:

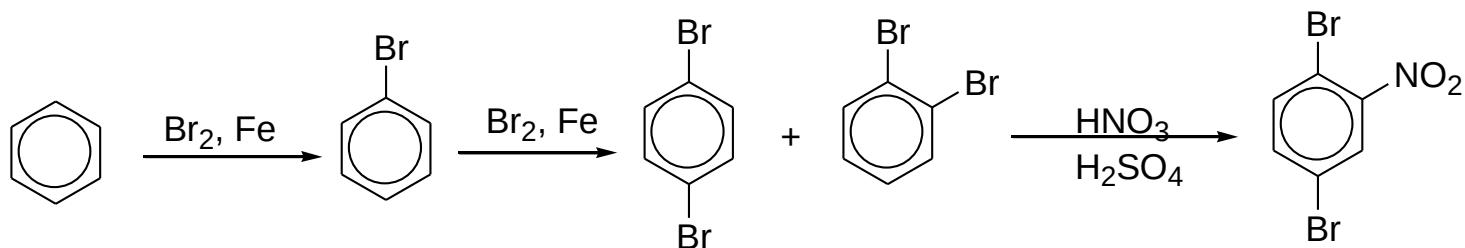


synthesis of *p*-bromonitrobenzene from benzene:

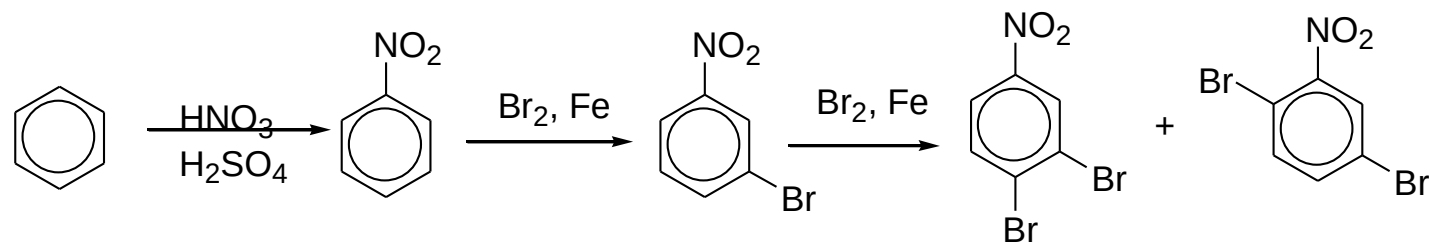


You may assume that you can separate a pure *para*-isomer from an *ortho*-/*para*- mixture.

synthesis of 1,4-dibromo-2-nitrobenzene from benzene



separate pure para isomer from ortho/para mixture



cannot assume that these can be separated!